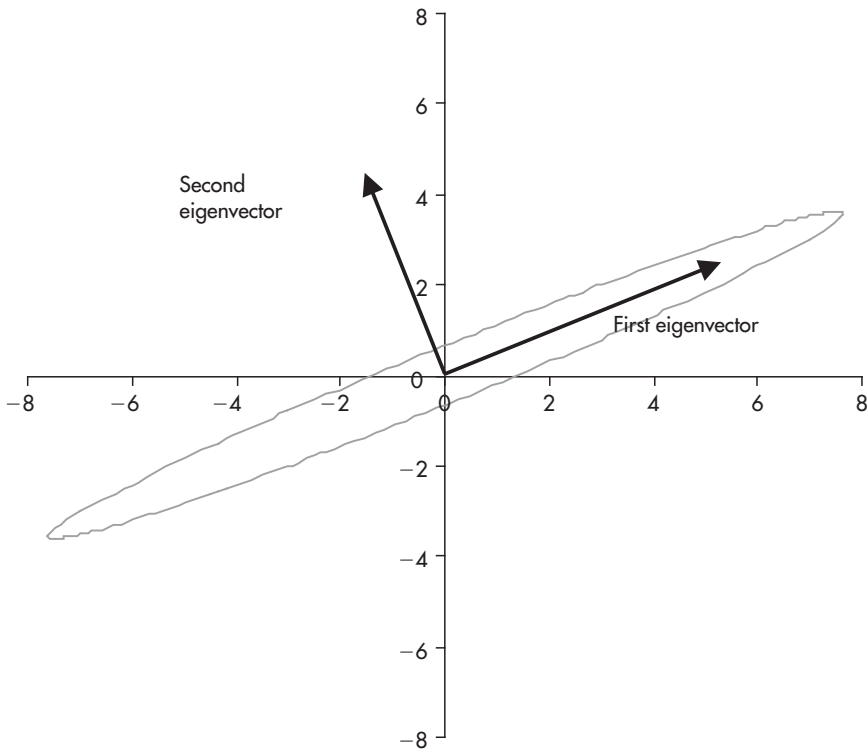




**FIGURE 3.2** Cov and examples for its two eigenvectors.

Source: Authors.

### Technical Point: Numerical Issues Involved in Eigenvalue Calculation

Eigenvalue calculation is famous for its numerical challenges. The main problem is to distinguish between two different eigenvalues and two slightly different numerical representations of the same eigenvalue. Since some of the eigenvalues tend to get very small when their number increases, this distinguishing becomes quite difficult for large covariance matrices and requires numerical representations with many digits and very high accuracy.

When an in-house PCA tool is required, the following hints could be useful:

- Scale the covariance matrix appropriately.
- Use a combination of numerical methods. In particular, since the largest eigenvalue is of crucial importance for the application of PCA to markets, confirm it through the Lanczos algorithm.

- Use as starting points for a Newton algorithm on the characteristic polynomial of the covariance matrix after a Householder transformation the following set of numbers:  $a + 2 \cdot b \cdot \cos\left(i \cdot \frac{\pi}{n+1}\right)$  for  $i = 1, \dots, n$ , with  $a = \frac{1}{n} \sum_{i=1}^n a_i$ ,  $b = \frac{1}{n-1} \sum_{i=2}^n b_i$  and with  $\begin{pmatrix} a_1 & b_2 & 0 & 0 \\ b_2 & a_2 & \ddots & 0 \\ 0 & \ddots & \ddots & b_n \\ 0 & 0 & b_n & a_n \end{pmatrix}$

being the Householder transformation of the covariance matrix.

- Additionally, run a systematic search for eigenvalues, whereby the largest eigenvalue found via the Lanczos algorithm determines the width of steps and range of starting points for the Newton algorithm.
- Run checks, whether the computed eigenvectors are really eigenvectors (i.e. fulfill  $Ax = \lambda x$ ) and orthogonal to each other.

The PCA sheet on the website accompanying this book implements these tricks. However, also due to numerical restrictions of Excel, it reaches its limitations rather soon and should not be expected to work for larger covariance matrices.

## PCA AS A FACTOR MODEL

PCA becomes a factor model by using the eigenvectors  $e_{ij}$  of the covariance matrix as factor loadings, that is by defining  $f_{ij} := e_{ij}$ , with the factor model being thus:

$$\begin{pmatrix} y_1^t \\ \vdots \\ y_n^t \end{pmatrix} = \sum_{i=1}^n \alpha_i^t \cdot \begin{pmatrix} e_{i1} \\ \vdots \\ e_{in} \end{pmatrix}$$

where  $\begin{pmatrix} e_{i1} \\ \vdots \\ e_{in} \end{pmatrix}$  is the  $i$ -th eigenvector.

We sort the eigenvalues (and associated eigenvectors) by the percentage of total variation explained, that is

$$|\lambda_1| \geq |\lambda_2| \geq \dots \geq |\lambda_n|$$

Hence, the first factor explains most of the market variation, the second explains most of the market variation not explained by the first factor, and so on. Correspondingly, the first factor loading, that is the first eigenvector, reveals the structure of the most important market mechanism, the second eigenvector the structure of the second most important, and so on. The importance of a market mechanism, that is the strength of its impact on the overall market variation, is quantified by the eigenvalues.

### Technical Point

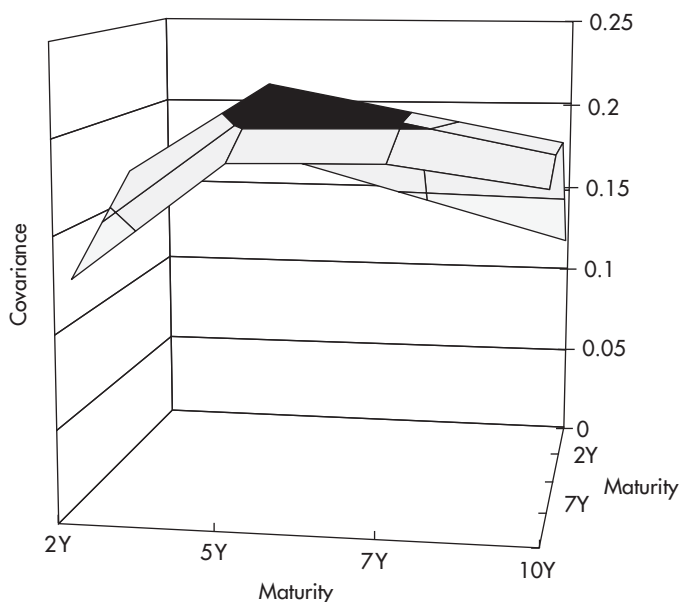
If  $x$  is an eigenvector of  $Cov$ , then for every  $a \neq 0$ ,  $ax$  is also an eigenvector of  $Cov$ . Thus, for every eigenvalue, there exist infinitely many eigenvectors, pointing in the same direction (or, if  $a < 0$ , in the opposite direction), but of different length. Any one of those can be chosen as a factor loading, and there is no reason to prefer one over the other.

In the end, one needs to arbitrarily decide the length of a particular eigenvector. For example, eigenvectors often are scaled to have unit length. However, there still exist two eigenvectors,  $x$  and  $y$ , with length one, with  $x = -y$ . Again, there is no criterion to decide for one or the other and one has to arbitrarily choose one. However, it is therefore important for the analyst to conduct the analysis and interpret the results with the particular choices of eigenvectors in mind. For example, if rates are an increasing function of factor 1 when  $x$  is chosen as the eigenvector, then rates will be a decreasing function of factor 1 when  $y = -x$  is chosen as the eigenvector. Many analysts have fallen into this trap, which can be avoided only by constantly remembering the eigenvectors. As in this example, factors are only meaningful when analyzed in conjunction with the factor loadings.

## INSIGHT INTO MARKET MECHANISMS THROUGH INTERPRETATION OF THE EIGENVECTORS

So far, we have developed the mathematical framework for PCA. For the rest of the chapter, we will see how PCA reveals the inner structural relationships of the market and how it can be applied to trading.

For the following example, we use weekly data from 4 Jan 2010 to 3 Oct 2011 for generic Bund yields (yield level, not change) for two, five, seven and 10 years. The input data and the PCA can be seen in the PCA sheet accompanying this book. In the first step, the covariance matrix is calculated and depicted in Figure 3.3. While this is not a necessary step in the analytical process based



**FIGURE 3.3** Covariance across the Bund yield curve.

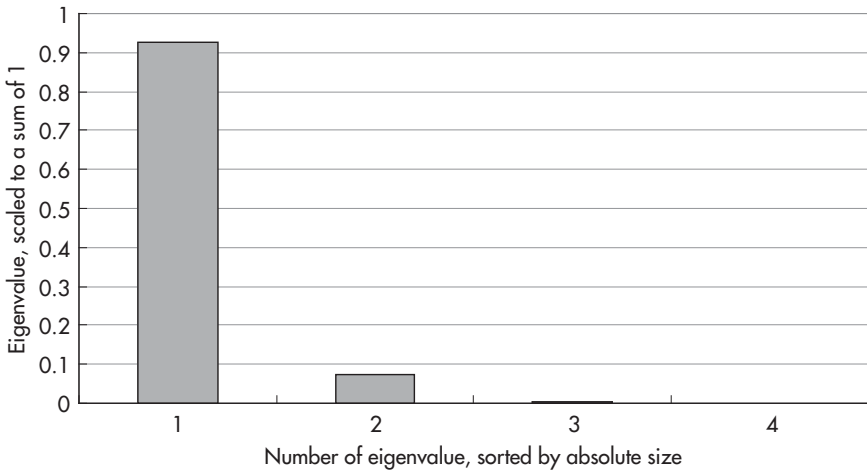
Sources: data – Bloomberg; chart – Authors.

Data period: 4 Jan 2010 to 3 Oct 2011, weekly data.

on PCA, displaying and examining the covariance matrix (i.e. the input into a PCA) can already give an intuition about the market mechanisms (i.e. the output of a PCA). In Figure 3.3, we can observe an area of maximal covariance in the medium part of the yield curve.

Then, the information about the market (as represented in the covariance matrix) is extracted through a PCA. As with any factor model, the eigenvectors of the covariance matrix represent the structural relations and mechanisms between the data series, while the eigenvalues show the relative significance of these factors in explaining the behavior of the data.

A typical situation for the relative strength of the factors, as quantified by the relative size of their eigenvalues, is depicted in Figure 3.4. As can be seen in Figure 3.4 in the case of the Bund yield curve, the first factor explains more than 90% of the yield curve variation; and the first three factors together, almost everything. This means that basically the whole information the Bund market provides can be reduced to, captured, and expressed in three numbers (the first three factors), with the factor loadings translating the information between the full Bund market and the three factors (numbers) back and forth.



**FIGURE 3.4** Scaled eigenvalues of a PCA on the Bund yield curve.

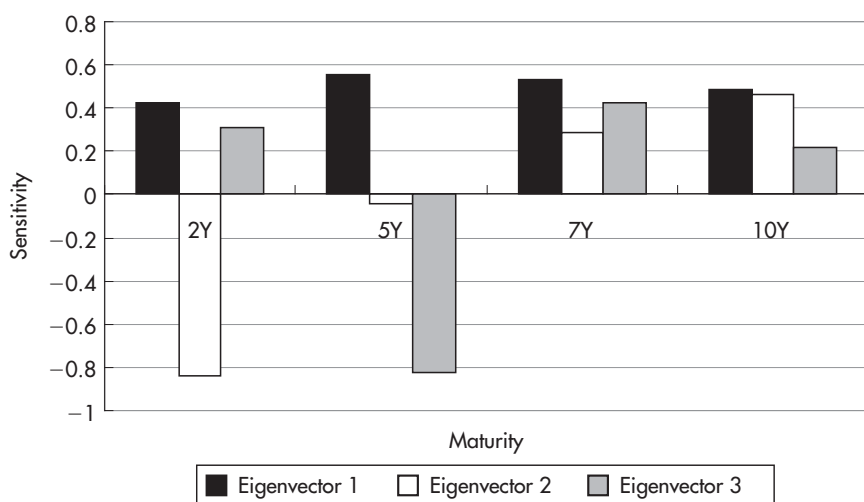
Sources: data – Bloomberg; chart – Authors.

Data period: 4 Jan 2010 to 3 Oct 2011, weekly data.

The next step to extract information about market mechanisms from a PCA is to examine the shape of the eigenvectors. The interpretation of the structural information contained in the eigenvectors can be done by applying the following scheme: if the  $i$ -th factor ( $\alpha_i$ ) increases by one, what happens to the rate curve? The answer to this question translates the market mechanisms from the mathematical language of eigenvectors of the covariance matrix into everyday terms.

In the following, we provide this interpretation for the first three eigenvectors of a PCA on the Bund curve, which are depicted in Figure 3.5. In this example, we find that a unit increase in the first factor corresponds to an increase in every point along the yield curve, since every entry in the first eigenvector has the same sign. Hence, we can interpret the first factor as representing the directional dynamics of the yield curve.

Moreover, the *shape* of the first eigenvector in Figure 3.5 represents the *shape* of directional moves: if  $\alpha_1$  increases by one, all yields increase, but medium yields (five-year, or 5Y) increase more than both short (2Y) and long (10Y) yields. This can be translated into everyday terms by saying that the pivotal point of directional moves of the yield curve is the 5Y area. We note that the output (eigenvector) of a PCA corresponds to the input (covariance matrix), with the shape of the first eigenvector reflecting the maximal covariance in the medium part of the yield curve we observed in Figure 3.3.



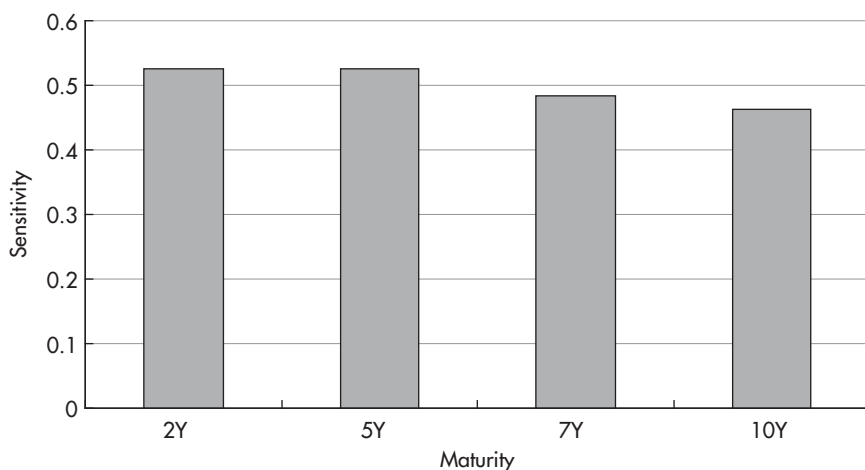
**FIGURE 3.5** First three eigenvectors of a PCA on the Bund curve.

Sources: data – Bloomberg; chart – Authors.

Data period: 4 Jan 2010 to 3 Oct 2011, weekly data.

Now, let's compare this with the first eigenvector of a PCA on the Bund curve with data from 1993 to 1997, which is depicted in Figure 3.6. While for 5Y, 7Y, and 10Y the sensitivity is almost identical to the current situation, back in the 1990s, the sensitivity of 2Y yields was as high as that of 5Y yields (i.e. the market exhibited a bull-steepening/bear-flattening pattern). This is typical for a market with an active central bank driving the yield curve up and down, which corresponds to a higher sensitivity of shorter yields to directional moves, hence a higher entry in the first eigenvector. Thus, the decreasing activity of the central bank from the 1990s until 2011 (partly a function of rates approaching zero) is reflected by the decreasing sensitivity of 2Y yields to the first factor. Note how the mathematics (entry in eigenvector) corresponds to real economic mechanisms. Even if we did not know anything about central bank history, one look at the PCA results would reveal both the structural shift in market mechanisms (pivotal point of directional moves shifting from the short end to the medium curve sector) and its likely source. Also note the stability displayed by the eigenvector over 20 years (apart from the short end), which we shall analyze in more detail later in this chapter.

Furthermore, note the difference to the approach of a factor model in the first category, like the NS model: rather than *assuming a priori* that direction is the strongest market dynamic, the market reveals that this is the case for this



**FIGURE 3.6** First eigenvector of a PCA on the Bund curve with data from 1993 to 1997.

Sources: data – Bloomberg; chart – Authors.

Data period: 1 Jan 1993 to 31 Dec 1997, weekly data.

particular market segment. And contrary to the *a priori* assumption of parallel shifts in the NS model, it shows that the real directional market mechanism follows a 2Y-5Y bear-steepening/5Y-10Y bear-flattening shape.

Also, together with the information about the eigenvalues shown in Figure 3.4, we have found (rather than assumed) that directional market mechanisms have indeed the strongest impact on the yield curve and can quantify their strength relative to other factors. Moreover, we could compare the strength of directional mechanisms in different markets and how it evolves over time by following the evolution of the scaled first eigenvalue. Interestingly, we found that the development of markets is characterized by a decreasing scaled first eigenvalue. In undeveloped markets, almost 100% of the action is explained by one single variable (direction), while increasing sophistication results in other mechanisms gaining strength. For example, while around the year 2000 the Indian domestic bond market had a scaled first eigenvalue of virtually one, it has now decreased to levels more in line with Western government bond markets, reflecting its increasing development. The Indian MIFOR–MIBOR basis swap market, on the other hand, has always been similar to the Bund market depicted in Figure 3.4. This could be interpreted as evidence that the basis swap market has induced the development of the domestic Indian bond market. In a sense, the scaled first eigenvalue is a universal (reverse) indicator of the sophistication of a market, allowing comparisons over space and time.

Moving on with the interpretation to the second eigenvector, if  $\alpha_2$  increases by one, short yields decrease and long yields increase. Thus, the second factor represents the slope element of the curve that is not explained by the first factor (i.e. by directional impacts on the slope). This is due to the entries in the second eigenvector crossing the x-axis once. Again, the shape of the second eigenvector reveals and quantifies the shape of steepening moves.

And if  $\alpha_3$  increases by one, short yields increase, medium yields decrease, and long yields increase, which we interpret as the curvature dynamics of the curve (not explained by the first and second factors). This corresponds to the entries in the third eigenvector crossing the x-axis twice.

These results are typical, with the  $i$ -th eigenvector crossing the x-axis  $i - 1$  times.<sup>6</sup> However, it is not always the case, as in a particular market steepening moves could explain more of the overall yield curve variation than directional moves. Sometimes, no reasonable interpretation is possible at all, which could indicate that modeling this particular market through PCA is not useful. An advantage of PCA versus factor models falling into the first category is that it reveals these issues as they are, including cautioning against its own use when appropriate.

## APPLYING EIGENVECTOR INTERPRETATION IN DIFFERENT MARKETS

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Statistical models like PCA require no specific knowledge about the instrument that is being modeled and hence are universally applicable. PCA only needs to know the time series, not whether the time series represents yields, swap spreads, or volatilities, or what drives that time series. Thus, the range of its possible applications is far larger than the yield curve example we have been using so far. In fact, interpretation of eigenvectors may become less predictable and straightforward, hence more interesting and revealing, when PCA is applied to other markets. Here, we illustrate the use of PCA in a number of different contexts, starting with volatility data as input variables.

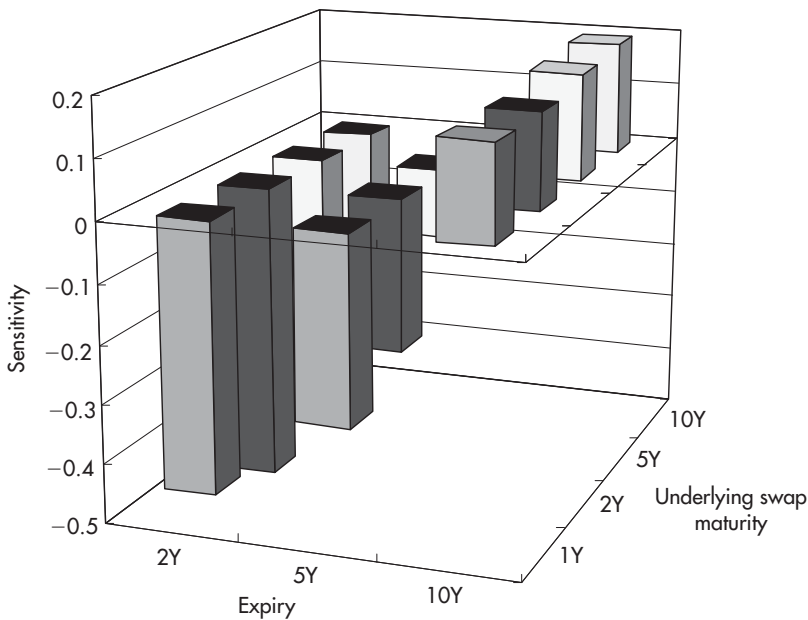
The two-dimensional surface of at-the-money-forward (ATMF) volatilities (or, if skew is considered as well, the three-dimensional volatility cube) must first be transformed into a one-dimensional vector. After the PCA is conducted, the outcome in vector form can be displayed again in a two-dimensional format. The complete results are discussed in detail in Chapter 19, from which we pick the chart of the second eigenvector as a typical result in Figure 3.7.

Usually, the first factor represents the overall level of volatility, the second factor differentiates across expiry, and the third one across underlying swap

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<sup>6</sup>One can back up this statement theoretically by linking PCA to a Fourier analysis.





**FIGURE 3.7** Second eigenvector of a PCA of the vega sector of the JPY implied volatility surface.

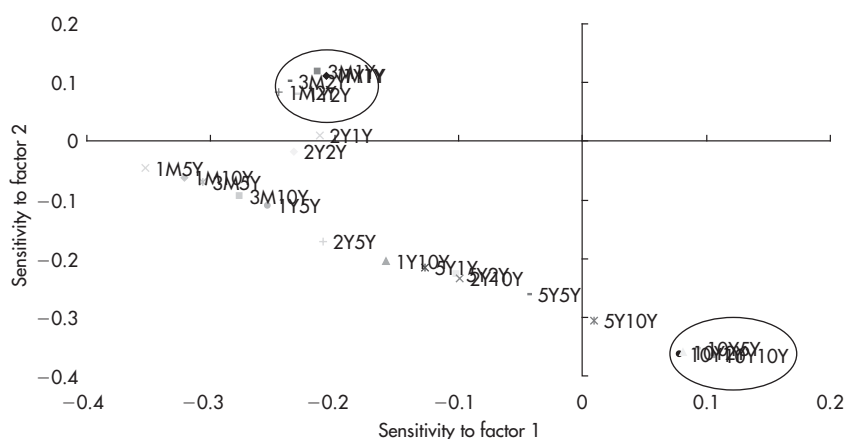
Sources: data – Bloomberg; chart – Authors.

Data period: 5 Jan 2009 to 19 Sep 2011, weekly data.

maturity of the options. However, the second and third factors sometimes change place (i.e. the differentiation across underlying maturity explains more of the overall variation of the volatility surface than the differentiation across expiry). Thus, PCA decomposes the volatility surface into its two dimensions (expiry and underlying swap maturity), with the first factor affecting both dimensions and the second and third factors reflecting the dimension-specific information.

When the number of instruments in the data set increases, as in the case of volatility analysis, the ability of PCA to reduce the dimensionality of the data becomes increasingly important. One can use that ability to detect relationships in large data sets. One tool supporting this process is cluster analysis, which depicts each instrument as a function of its sensitivity to various factors (typically the first and second, as in Figure 3.8<sup>7</sup>). If the first and second

<sup>7</sup>Note that the input data for Figure 3.8 consist of the whole volatility surface, while for Figure 3.7 only options with an expiry of at least 2Y are used. The reason for this is explained in Chapter 19. As there are two different PCAs behind the two charts, the sensitivities to factor 2 displayed in Figure 3.7 and Figure 3.8 are also different.



**FIGURE 3.8** Example for a cluster analysis of the whole JPY volatility surface.

Sources: data – Bloomberg; chart – Authors.

Data period: 5 Jan 2009 to 19 Sep 2011, weekly data.

eigenvalues are large relative to the others, then the behavior of an instrument will be largely determined by its sensitivity to the first and second factors. Thus, if two instruments have similar sensitivities to the first and second factors (i.e. they are close to each other in Figure 3.8), they can be expected to behave similarly (i.e. to form a “cluster”). In the example in Figure 3.8, we can observe two clusters: one containing all options with both short expiries and short underlying swap maturities and the other containing all options with both long expiries and long underlying swap maturities. We therefore conclude that volatilities at both ends of the diagonal of the volatility surface usually move closely together, while further away from those two corners of the volatility surface, options behave more individually. The same sort of analysis can be applied to other combinations and numbers of factors as well, for example, by forming three-dimensional clusters of the first three sensitivities.

In Chapter 13, we shall use PCA to gain insights into the structure of various credit default swap (CDS) markets. Picking one result as an example, Figure 3.9 shows the first three eigenvectors of a PCA with the 5Y CDS quotes of sovereign issuers in the core Eurozone as input. The first eigenvector (in which all entries are positive) represents the overall level of Eurozone CDS quotes, with the sensitivities measuring the impact a general widening of Eurozone CDS quotes has on individual countries: it affects France more than Austria, Austria more than the Netherlands, and the Netherlands more